

Talent Carriage

IMAGINE INNOVATE TRANSFORM

PRODUCT SCOPE DOCUMENT · VERSION 1.0

AI Attendance Agent

An AI agent that closes attendance gaps by itself — it talks to the employee on WhatsApp or by phone, answers their questions about company policy, gets the manager's approval, and completes the action inside the customer's HRMS.

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DOCUMENT TYPE

Functional & technical scope

STATUS

For review and sign-off

PRODUCT MODEL

Multi-tenant SaaS, sold to employer customers

CLOUD

Microsoft Azure

What this document is. Everything the platform will do, how it adapts to each customer's HRMS and attendance codes, the technology proposed, and the order of delivery. Items marked "prospective" may change during build; the functional scope is what we are agreeing to.

Where we are today. A working demo covers the first stage: it reads an attendance register, finds absentees and half-days, sends the WhatsApp question, understands replies — including free text in English and Hinglish — and responds with the correct instruction.

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Sections 1 to 3 need no technical background. Sections 4 to 10 carry the detail an IT team or HRMS vendor will ask for. Sections 13 and 14 are the practical ones: what each customer must provide, and when each capability is ready. Phase labels show delivery order: **PHASE 1** core platform · **PHASE 2** mapping & integration · **PHASE 3** actions & approvals · **LATER** voice, analytics, scale-up.

1. The product in one page

A product Talent Carriage sells to employers. It watches attendance data, talks to the employees whose attendance is incomplete, and finishes the paperwork for them.

1.1 The problem it removes

Every month, HR teams spend days chasing the same thing: employees marked absent, on half-day, or with a missed punch, who have not applied for leave or regularisation. Payroll cannot close until each one is explained and approved. Today it is done with phone calls, hand-typed WhatsApp messages and spreadsheets — slow, repetitive, and badly recorded.

1.2 What the agent does instead

- **It notices.** Each day, at a time the customer sets, it reads that day's attendance and finds every gap: absent, half-day, missed punch, or any other status that customer treats as needing an explanation.
- **It asks.** One WhatsApp message per gap, in the employee's language, with buttons to reply.
- **It talks.** Employees can reply with a button, a number, or plain typed text — and can ask questions back: how much leave they have left, what the regularisation deadline is, what to do if they were at a client site. The agent answers

from that employer's own policy.

- **It acts.** This is what makes it an agent rather than a reminder tool. If leave or regularisation needs applying, it asks the manager to approve on WhatsApp and then performs the action in the customer's HRMS.
- **It chases.** Still pending after two days? A reminder. Still pending on day three? An automated phone call that holds a short spoken conversation and records the outcome.
- **It reports.** Every case, message, approval and action, with a full audit trail and clean month-end numbers.

1.3 Who uses it

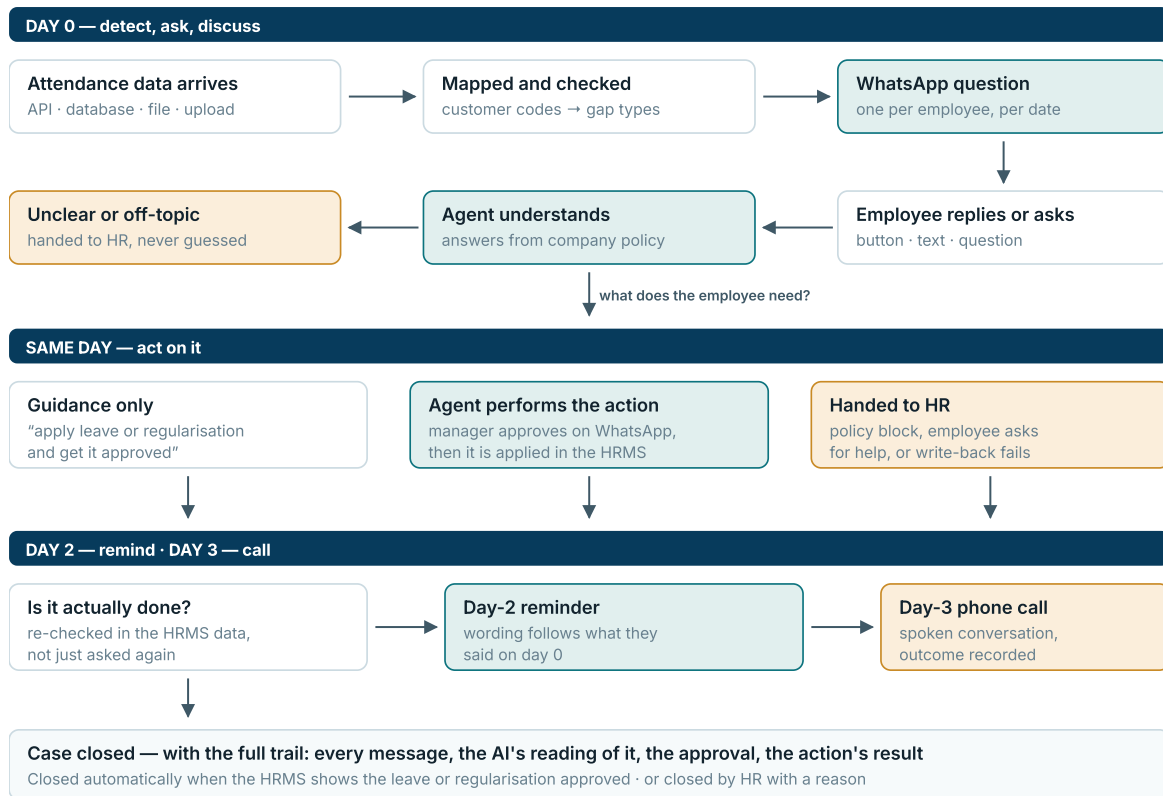
USER	WHERE	WHAT THEY DO
Employee	WhatsApp and phone	Answers the question about their attendance, asks policy questions, asks for leave or regularisation to be applied for them.
Manager	WhatsApp, or the portal	Approves or rejects with one tap; sees what is pending with their team.
HR user	Web portal	Watches the day's cases, handles what the agent flags, runs month-end reports.
HR admin (customer)	Web portal	Sets the rules: timings, which statuses to chase, message wording, escalation, policy content.
Talent Carriage team	Admin console	Onboards customers, maps their data, connects the HRMS, monitors health across all customers.

1.4 Why it can be sold to many different employers

No two employers record attendance the same way. One writes **A|A** for a full-day absence and **P|A** for a second-half absence; another writes **Ab**, **CL**, **SL**. One stores *Full name*, another *First name* and *Last name*. One has a modern HRMS with an API, another exports an Excel file daily. The platform therefore keeps everything customer-specific in a **mapping layer**, configured once during onboarding. Detection, messaging, AI, actions and reporting are then identical for every customer — one product, sold repeatedly, not rebuilt per employer. Section 3 explains it.

2. How it works, end to end

One employee, one date with a gap, followed until it is closed — the flow from the agreed process flowchart, with conversation and action steps added.



Green boxes are agent actions; amber boxes need a human. Every timing shown is a per-customer setting.

Day 0. The platform pulls the day's attendance, translates it through that customer's mapping, and creates **cases** — one case is one employee and one date. The employee gets a single, clearly dated WhatsApp message with reply buttons. Three open dates means three separate conversations, never a confusing bundle.

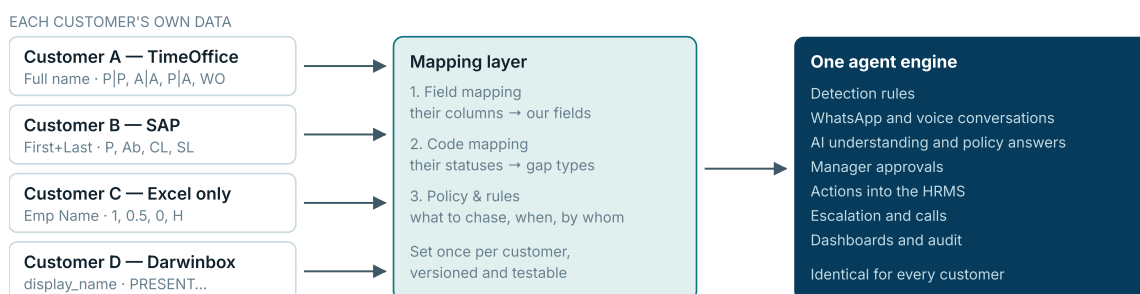
Reply, discussion and action. The four standard answers from the agreed flow (*I was absent, I was working, already applied leave, already sent regularisation*) each get the correct instruction immediately. The employee can also ask questions, and where the customer has enabled actions the agent offers to do the work: "Shall I apply casual leave for 18-Aug and send it to Deepak for approval?"

Day 2. Before any reminder, the HRMS is re-checked. If the leave was applied and approved meanwhile, the case closes quietly — no nagging. Otherwise the reminder is worded for that person's situation.

Day 3. Cases still open are called. The call is a short spoken conversation, not a recorded announcement: the employee can say what happened, ask a question, or ask for HR. Outcomes route to HR with a recommended next step.

3. Built to fit any customer

The heart of the product. Each customer's data differs; the agent's logic must not. Three kinds of mapping make that possible, all done by filling in a screen rather than writing code.



Everything customer-specific lives in the mapping layer. The engine itself never changes.

3.1 Field mapping — “where is the employee's name?”

The platform needs a small set of facts. The customer's own column names are pointed at them once. Names can be built from several columns, and values cleaned on the way in (trim spaces, add a country code, uppercase a code).

WHAT THE PLATFORM NEEDS	REQUIRED	WHAT CUSTOMERS ACTUALLY HAVE
employee_code	Yes	Employee Code , EmpID , PERNR , staff_no
employee_name	Yes	Full name , or First Name + Last Name , or display_name
mobile	Yes	Mobile number , Contact , phone_primary — with or without +91
date	Yes	a column per day (1 / 8 Sat), or one row per day (att_date)
attendance_status	Yes	P A , Ab , 0.5 , PRESENT , or in/out punch times
manager_identifier	Yes*	Reporting manager , manager_emp_code , L1 Manager Email
department, branch, shift, location	Optional	routing, reporting and working-hour rules
date_of_joining, employment_status	Optional	skip new joiners, exits, and people on long leave

* Needed for approvals. A manager's *name* cannot receive a WhatsApp — a manager mobile, or a manager code that resolves to one, is required. Where a customer cannot supply it, approvals fall back to a nominated HR approver.

3.2 Attendance-code mapping — “what does this code mean?”

The platform thinks in a fixed internal vocabulary (Appendix A). Each customer's codes map onto it, and the mapping says which meanings are a gap worth chasing, which are informational, and which close a case by themselves.

CUSTOMER'S CODE	PLATFORM MEANING	CHASE IT?	NOTES
A A	Absent — full day	Yes	Two-session format: first half second half
P A / A P	Absent — half day	Yes	The message names the exact half
Ab , ABS , 0	Absent — full day	Yes	Single-code format
0.5 , HD	Absent — half day	Yes	Which half may come from another column
MP , MIS	Missed punch	Yes	Usually needs regularisation, not leave
LT , EG	Late in / early out	Customer's choice	Often chased only past a threshold
CL , SL , EL	Leave — approved	No	Explained; closes an open case
LWP , LOP	Leave without pay	Customer's choice	Some employers still want confirmation
CL-P , PEND	Leave — pending approval	Manager chased	The manager is the blocker, not the employee
WFH , OD , TR	Working elsewhere	No	Present for our purposes
WO , HO , H	Weekly off / holiday	Never	Excluded before anything runs
anything unmapped	Unknown	Never	Never messaged; reported to the admin to map

DELIBERATE SAFETY RULE

An unmapped code never triggers a message. Silence is safe; guessing is not. The admin screen shows a live count of unmapped codes from the last import, so gaps surface immediately.

3.3 Policy and rules — “how does this employer work?”

Per-customer settings, with sensible defaults, most also settable per department or location: daily check time and time zone · which meanings to chase · grace periods and thresholds · quiet hours and working days · holiday and shift calendars · who to exclude (new joiners, exits, long leave, named roles) · escalation timings (reminder day, call day, retries) · approval chain and timeouts · languages offered · which actions the agent may perform and which leave types it may apply · message wording and signature. Alongside these sits the **policy pack**: the customer's attendance, leave and regularisation rules in their own words, used to answer employees' questions (section 6.3).

3.4 The mapping studio

1. **Bring a sample.** Upload a real file, or connect the HRMS and pull a few days.
2. **Auto-detect.** The platform lists the columns and the distinct status codes found, with counts, and AI proposes what each one means. A human confirms.
3. **Fill the gaps.** Anything unmapped is highlighted and must be decided before go-live.
4. **Dry run.** It shows exactly who *would* have been messaged for a past date, and the exact wording, with nothing sent. This is how a customer signs off the configuration.
5. **Save as a profile.** Versioned with an audit trail, and clonable as the starting point for the next customer on the same HRMS — which is what shortens each onboarding to a review.

4. Functional scope by module

#	MODULE	CAPABILITIES	PHASE
M1	Customer (tenant) management	Create and configure a customer; branding; licensed headcount; test and live modes; suspend or offboard with data export	1
M2	Users, roles and access	Platform admin, HR admin, HR user, manager, auditor; invitations; SSO for customer staff; department scoping; access logs	1
M3	Employee master	Codes, mobiles, managers, department, location, shift, joining and exit dates; sync from HRMS or file; change history; consent and opt-out state	1
M4	Data ingestion	Manual upload; scheduled file pick-up (SFTP or Blob); HRMS API pull; read-only database access; HRMS push; per-run import report with rejects; safe re-import	1 2
M5	Mapping studio	Field and code mapping, transformations, AI suggestions, unmapped-code alerts, versioning, profile library, dry runs against past dates	2
M6	Detection engine	Applies mapping and rules to create cases; calendars, shifts, grace and exclusions; one case per employee per date; never duplicates; safe re-runs; backdated corrections	1
M7	Conversation engine	WhatsApp send and receive; template management; buttons and menus; multi-language; 24-hour window handling; delivery receipts; retries; opt-out; full transcript per case	1
M8	AI understanding and dialogue	Intent and detail extraction from any reply; multilingual including Hinglish; policy questions answered from the customer's policy pack; confidence thresholds; handover instead of guessing; every decision logged	1 3
M9	Action engine	Real HRMS work: apply leave, apply regularisation, approve, cancel, raise a ticket; manager approval loop; eligibility pre-checks; idempotent and retried; receipts; fallbacks (section 5)	3

M10	Escalation and voice	Day-2 reminders worded by original answer; day-3 conversational call with keypad fallback; retries and voicemail; escalation to manager, skip-level and HR; SLA clocks	3 LATER
M11	HR workspace	Daily dashboard; case list with filters and search; case detail with transcript and audit; manual send, resend, close with reason, reassign; bulk actions; AI-flagged queue; exports	1
M12	Manager experience	Approve or reject on WhatsApp; pending-items digest; a light web view without full login; delegation while on leave	3
M13	Scheduling and automation	Per-customer daily run; catch-up after a failed import; quiet hours; month-end sweep; retry queues; manual run-now; run history	1
M14	Reporting, audit and admin	Operational and month-end reports; AI accuracy review; full audit trail; scheduled email packs; platform-wide health; usage metering for billing	2 LATER

The demo already covers a slice of M4, M6, M7, M8 and M11 for one customer. The product work is to make it multi-customer, mappable, conversational and action-capable.

5. The Action Engine

The difference between a reminder tool and an agent: the employee asks, the manager approves, and the platform performs the action in the HRMS and proves it did.

Employee: taps *Yes, I was absent*

Agent: "You have 6 casual leave days available. Shall I apply casual leave for 18-Aug-2026 and send it to Deepak for approval?" [Yes, apply it] [I'll do it myself] [Different leave type]

Employee: taps *Yes, apply it*

Agent → Manager: "Ankit Panwar has requested casual leave for 18-Aug-2026. Reason: fever. Balance after approval: 5 days." [Approve] [Reject] [Ask for details]

Manager: taps *Approve*

Agent: applies it in the HRMS, then confirms to both with the HRMS reference. Case closed.

5.1 How an action is defined

Actions are definitions the platform executes, not one-off code, so new ones can be added later without redesign. Each has a **trigger** (an intent, a case state, or an HR click), **inputs** (date, half or full, leave type, reason), **pre-checks** (balance, backdating window, overlaps, probation, blackout dates), an **approval** step (who, in what order, with what timeout and fallback), **execution** (the connector call or a fallback route), **proof** (the HRMS reference stored on the case), **failure handling** and **notifications**. The v1 catalogue is in Appendix A.

5.2 Rules the engine holds itself to

- **Never act without a mandate.** An explicit employee request, plus an explicit approval where configured. An ambiguous reply never results in a write.
- **Never act twice.** Every action carries an idempotency key, so a retry, a duplicate WhatsApp delivery or a double tap cannot create two leave requests.
- **Check before writing.** Balance, eligibility and duplicate checks run first, so the employee is told why in plain language instead of a failure appearing later.
- **Prove it.** The HRMS's own reference is stored and shown. If nothing verifiable comes back, the action is marked *unconfirmed* and read back on the next sync rather than assumed successful.
- **Fail loudly, to a person.** A failed write retries, then lands in an HR queue with the error and the suggested fix. Never silently dropped.

5.3 When the HRMS cannot be written to

Some HRMS products have no write API, or IT will not allow one initially. The conversation and approval flow stay the same, ending in one of these instead: a daily **approved-request file** in the HRMS's import format · a **structured ticket or email** to HR with the approval attached · an **HR action queue** in our portal where HR performs it and confirms · or **guidance only**, with completion verified from the next data sync.

6. WhatsApp: a real conversation

WhatsApp is the employee's interface: no app to install, no login, no training. Beyond the buttons, the agent holds a genuine conversation about attendance and the employer's policies.

6.1 The channel mechanics

- **The first message must be a Meta-approved template.** Each customer's templates are submitted and tracked in the platform, with approval status visible.
- **After the employee replies, we can talk freely for 24 hours.** All discussion, offers, approvals and confirmations happen inside that window; after it closes, the next contact is a template again.
- **Meta charges per conversation**, not per message (section 16). **Opt-out is permanent**, and those employees are routed to HR instead.
- **The sending number is the customer's own**, so employees see their employer. Either the customer's Meta account is connected, or — recommended for selling at scale — Talent Carriage becomes a Meta **Tech Provider** and the customer connects their number through a short guided flow in our portal.

6.2 What the conversation can cover

The agent is a subject-matter assistant for attendance, not a general chatbot. Within that subject it can:

EMPLOYEE SAYS	AGENT DOES
"I was on client site that day"	Recognises a regularisation case, confirms the date and reason, offers to apply it
"How many leaves do I have left?"	Reads the live balance from the HRMS and answers, by leave type
"Till when can I regularise August?"	Answers from that employer's policy pack, quoting the actual rule
"No casual leave left, what now?"	Explains the employer's unpaid-leave or regularisation route, then offers to start it
"Mera manager chhutti pe hai"	Applies the approval fallback: alternate approver or HR, per configuration
"Talk to HR"	Hands the case to HR with the transcript, and tells the employee who will follow up
Anything outside attendance (salary, appraisal, IT)	Politely declines and points to the right HR channel. It does not improvise.

6.3 Where the policy answers come from

Each customer's attendance, leave and regularisation policy is loaded as a **policy pack** — their own documents, plus short question-and-answer entries HR can edit in the portal. Answers are generated *only* from that pack and from live HRMS data, and the portal shows HR which clause each answer came from. If the pack does not cover a question, the agent says so and offers HR help rather than inventing a rule.

TO BE DESIGNED IN DETAIL

Exactly how policy content is ingested, indexed and kept current — and how much HR edits directly versus re-uploading a document — is a design exercise in Phase 3. The principles above (grounded answers, visible source, no invention, HR review before enabling) are fixed.

6.4 Guardrails

- **Approved wording starts every conversation;** generated text only inside a live conversation, and only on attendance topics.
- **No promises the platform cannot keep:** it never says leave is approved until the manager has approved and the HRMS confirms. No other employee's data, no salary or payroll advice, no disciplinary opinions.
- **Sensitive disclosures** (illness, bereavement, harassment) get a short, human reply and go to HR; medical detail is not probed. **Abuse or distress** ends the automated exchange and escalates.
- **Every generated message is logged** with the model used, and HR can review a sample before conversational mode is enabled for a customer.

7. Voice: a real conversation

On day 3, unresolved cases are called. The call is a spoken exchange driven by the same agent brain as WhatsApp — not a recorded announcement with a keypad menu.

7.1 How the call works

1. The platform places the call through the customer's voice provider, respecting quiet hours and retry limits.
2. The agent introduces itself, states the pending date, and asks what happened.
3. **Speech to text** transcribes the reply; the **agent brain** decides what to say next using the case, the HRMS data and the policy pack; **text to speech** speaks it back.
4. The employee can answer naturally, ask a question ("how many leaves do I have?"), ask for a repeat, or ask for HR — in which case the call is transferred or an HR callback is logged.
5. Where the employee prefers it or the line is poor, it falls back to **keypad** input ("press 1 if completed, 2 if not, 3 for HR"), so an outcome is always captured.
6. The outcome, transcript and — where the customer enables it and the call announces it — the recording are stored on the case.

Conversation length is capped: anything unresolved after two turns goes to HR rather than looping. Target response time between the employee finishing a sentence and the agent replying is under two seconds.

7.2 Voice providers

One internal interface, so the provider can differ per customer or country:

PROVIDER	NOTES
Plivo	Indian and global coverage, straightforward programmable voice with streaming audio for live conversation; competitive India rates
Exotel	Widely used by Indian HR and support teams; strong India compliance posture and local support
Azure Communication Services	Stays inside Azure and our existing billing; convenient where a customer mandates one cloud vendor
Twilio	Mature tooling; costs and Indian regulatory steps checked per customer

Speech recognition and synthesis are chosen per language (section 8): Sarvam AI for Indian languages and accents, with Azure or the telephony provider's engines for English. Automated outbound calling in India carries consent and do-not-disturb obligations, so consent is captured at onboarding, calls respect quiet hours, and an employee can ask never to be called again.

8. The AI layer and model strategy

Several models, each used where it is strongest, behind one internal interface — so a model can be swapped, priced or restricted per customer without touching the product.

8.1 Which model does what

JOB	MODEL	WHY
Button taps, "1"–"4", exact phrases	No model — rules	Exact, instant, free. Most replies never reach a model.
Reply classification, English and Hinglish	GPT-4o mini or Claude Haiku	Cheap and fast on short prompts; both evaluated on real replies and the better one set as default
Indian-language understanding (Hindi, Gujarati, Marathi, Tamil, Telugu, Bengali...)	Sarvam AI	Built for Indian languages and code-mixed text; Indian data residency; better on Indic scripts than general models
Policy questions and dialogue	GPT-4o mini / Claude Haiku , grounded on the customer's policy pack	Needs careful reading of policy text and instruction-following inside strict guardrails
Speech to text on calls	Sarvam AI speech recognition	Accuracy on Indian accents and languages, including mixed speech
Text to speech on calls	Sarvam AI Indic voices; Azure or the provider for English	Natural Indic voices; English handled well by existing engines
Mapping suggestions, HR summaries, digests	GPT-4o mini / Claude Haiku	Internal helpers where cost matters more than nuance
HR analytics assistant	Best available at the time	Read-only questions over that customer's own case data

LATER

8.2 The model gateway

- **One interface, many providers.** The product asks to "classify this reply" or "answer this question"; the gateway picks the model based on language, task and the customer's configuration.
- **Per-customer choice.** A customer who mandates Indian-only processing can be pinned to Sarvam and an Indian-region deployment; another may prefer a single vendor. No code change either way.
- **Automatic fallback.** If a provider is slow or erroring, the gateway retries on the alternate model; if all fail, the case goes to HR rather than staying silent.
- **Cost control.** Rules first, short prompts, small models, cached policy answers, per-customer spend caps and alerts.
- **Evaluation before switching.** A saved set of real replies and policy questions is re-run against any new model or prompt, and accuracy compared, before it becomes the default.
- **No training on customer data**, contractually with every provider; only the text needed for the decision is sent, never the employee database.

8.3 What the AI is not allowed to do

- **It does not decide who gets messaged.** That comes from the mapping and rules — exact, reviewable logic.
- **It does not write to the HRMS.** It proposes; the action engine validates and performs.
- **It does not invent policy or balances.** Those are read from the policy pack and the HRMS.
- **It does not guess.** Below the confidence threshold it asks one clarifying question, then hands over to a human with the original message attached.

Every AI decision is logged with the message, the intent, the confidence and the model version, and is reviewable in a queue where HR can correct it. Corrections become test cases. Accuracy, ask-again rate and handover rate are reported per customer, so the automation is measured rather than assumed.

9. Connecting to a customer's HRMS

Every customer's IT maturity differs. Several ways in and out, with identical behaviour whichever is used.

READING ATTENDANCE	HOW IT WORKS	BEST FOR
Manual upload	HR uploads the register export in the portal. Available from day one.	Pilots, small employers
Scheduled file	The HRMS drops a daily file on SFTP or Azure Blob; we pick it up, import and report rejects.	Most mid-size customers
HRMS API pull	We call the HRMS API on a schedule; credentials held in Azure Key Vault.	Modern HRMS products
Database read	A read-only user or replica, queried through a mapped view.	On-premise or legacy
HRMS push	The HRMS calls our authenticated API when attendance is finalised.	Same-day speed

Every import produces a report: rows read, employees matched, rows rejected with reasons, unmapped codes seen, and what changed since last time. Later corrections are picked up, and a case closes automatically if the correction explains the gap. Writing back uses the routes in section 5.3.

Connectors are small, standard pieces built to a fixed internal contract — *fetch attendance*, *fetch employees*, *fetch leave balance*, *apply leave*, *apply regularisation*, *approve*, *read status* — each declaring what its HRMS supports, so the agent only offers an employee what can actually be delivered. A generic file connector and a generic REST connector cover many customers through configuration alone; named connectors for common HRMS products are built as customers arrive and reused afterwards. Every connector has a sandbox mode that reads real data but writes nothing, and all calls are logged, rate-limited and retried, with alerts on repeated failure.

PRACTICAL ONBOARDING PATH

Start read-only, run in guidance-only mode for a week or two so the customer trusts the detection and messaging, then enable write-back actions once their IT is comfortable.

10. Architecture and tech stack

Prospective, and aligned with the existing Talent Carriage Azure stack so the team operates one familiar toolset. Choices may be refined during build; the functional scope does not depend on them.

LAYER	PROPOSED TECHNOLOGY	WHY
Web app, portal and API	Next.js (React, TypeScript) on Azure App Service (Linux), validation with Zod	Same stack and deployment as the existing Talent Carriage client hub
Database	Azure Database for PostgreSQL (Flexible Server) with Drizzle ORM	Matches the existing project; strong multi-tenant isolation and audit querying
Background work	Azure Functions (timer and queue triggers)	Daily checks, reminders, retries and syncs, independent of web traffic
Queues	Azure Service Bus	Each send, action and webhook event processed exactly once, with a dead-letter queue
Files	Azure Blob Storage	Uploaded registers, reports, policy documents, call recordings, drop folders

Secrets	Azure Key Vault	Per-customer WhatsApp tokens, HRMS credentials, model and voice keys — never in code
Identity	Microsoft Entra ID for our staff; email or SSO for customer users	Customer IT teams expect SSO; Entra is already in use
AI	Azure OpenAI (GPT-4o mini), Claude Haiku , Sarvam AI behind our model gateway	Right model per job, per language, per customer policy (section 8)
Messaging	Meta WhatsApp Cloud API , Talent Carriage as Tech Provider	Official channel; embedded sign-up makes each new customer quick to connect
Voice	Plivo or Exotel (also ACS, Twilio) with streaming audio	Live two-way conversation, India-ready, swappable per customer
Monitoring	Azure Monitor and Application Insights	Failed sends, stuck queues, connector errors and slow runs raise alerts
Delivery	GitHub Actions → App Service deployment slots	Same pipeline style as the client hub; staging, swap, rollback
Edge	Azure Front Door or Application Gateway with WAF	TLS, custom domains, protection for public webhook endpoints

One platform, many customers. Every record carries a customer identifier, enforced in the database with row-level security, so one customer's data cannot be reached through another's session — not merely filtered in application code. Credentials, configuration, mapping profiles, templates, calendars, policy packs and action permissions are all per customer. A large or regulated customer can be given a dedicated database without changing the application, and usage is metered per customer for billing and fair-use limits.

Environments. Development (synthetic data, sending disabled), staging (release testing and each customer's onboarding dry runs), production (with a per-customer test mode). Every environment has a sending switch, so a customer can be fully configured and verified in production before a single employee is contacted.

Reliability details that matter in practice. Webhook events are acknowledged instantly and processed from a queue, so no reply is lost during a deployment. Duplicate protection throughout: one case per employee per date, one message per case, one action per request, deduplicated on provider message identifiers. A missed daily run catches up rather than skipping a day. Sends are paced per WhatsApp number so a month-end spike does not trip Meta's rate limits.

11. Security, privacy and compliance

The platform holds employee phone numbers, attendance records and personal reasons for absence, and is designed for that responsibility.

- **Access.** Role-based: platform admin, HR admin, HR user, manager, auditor. Managers see only their team; HR users can be scoped by department or location. SSO for customer staff; enforced multi-factor authentication for our staff. Our support access to customer data is time-limited, reason-logged and visible to the customer.
- **Protection.** TLS in transit, platform encryption at rest, encrypted backups. Secrets only in Key Vault, rotatable without downtime. Signature verification on every inbound WhatsApp webhook. Phone numbers and message content masked in logs. Dependency scanning, and an external penetration test before the first production customer.
- **Privacy roles and data.** The customer is the data fiduciary; Talent Carriage processes on their instructions under a data processing agreement. Only the fields in section 3.1 are ingested — salary, bank and identity data are never requested.
- **Notice and consent.** Onboarding includes an employee communication pack — what the agent is, what it sends, how to opt out of messages and calls. Consent and opt-out states are stored per employee.

- **Retention and residency.** Configurable per customer; 24 months by default for cases, transcripts and recordings, then deleted or anonymised. Offboarding produces a full export followed by deletion. Data and AI processing stay in Indian regions by default, including the Indian-language models.
- **Employee rights.** Access, correction and erasure requests are supported through the customer's HR, with tooling to find and export everything held about one person.
- **Channel rules.** Approved templates only to start conversations; opt-outs permanent; no marketing; quiet hours on messages and calls; recording only where enabled and announced.
- **Audit trail.** Each case keeps an unalterable history: the import row that created it, every message with delivery status, the AI's reading, confidence and model, each approval with who and when, each HRMS write with its reference or error, and every human intervention with a reason — exportable, and usable as evidence in a payroll dispute.

12. Reporting, scale and reliability

12.1 What HR and leadership see

Daily: gaps found, messaged, replied, resolved, awaiting manager, needing HR; a work queue of unclear replies, failed sends, failed HRMS writes and missing mobile numbers; case search with full transcripts. **Month-end:** the unexplained-absence register payroll needs, actions completed with HRMS references, items pending with managers ranked by age, exports to Excel and CSV, and a scheduled email pack. **Leadership:** reply rate and time to first reply, auto-resolution rate, average time to close, actions performed by the agent, escalation rates, AI accuracy and handover rate, repeat-offender and department patterns, and estimated HR hours saved. **For our own team:** a console across all customers showing import health, volumes, failure rates, template states, connector errors and model spend.

12.2 Design targets

TARGET	DESIGN GOAL
Customers · employees	Up to 100 active customers; 200,000 employees in total; 25,000 for a single customer
Daily attendance rows	Up to 500,000 processed inside the nightly window
Cases on a peak day	Up to 20,000 across all customers (month-end corrections)
WhatsApp messages per day	Up to 60,000, paced within Meta's per-number limits
Response time	Within 5 seconds on WhatsApp; under 2 seconds per turn on a call
Daily check duration	A 25,000-employee customer completes within 15 minutes
Availability · recovery	99.5% monthly for the portal; point-in-time restore 7–35 days; recovery point under 15 minutes, recovery time under 4 hours

Nothing depends on a single machine staying up: work is queued, retried and traceable, so a failure delays a message rather than losing a case.

13. Onboarding a new customer

Selling repeatedly depends on onboarding being routine — roughly two weeks of elapsed time per customer once the platform is built.

STAGE	WHAT HAPPENS	WHO
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Day 1–2	Kick-off and data sample: attendance export, employee list, attendance codes with meanings, holiday calendar, policy documents	Us with customer HR
Day 2–4	Mapping and rules: fields, codes, timings, exclusions, languages, policy pack loaded	Our onboarding team
Day 3–6	WhatsApp: connect the customer's number, submit templates, wait for Meta approval	Customer IT with our guidance
Day 4–7	Integration: agree the data route and, if in scope, the write-back route. Read-only first.	Customer IT with us
Day 6–8	Dry run and sign-off: run over a past month with sending off, and review exactly who would have been contacted and what they would have received	Customer HR signs off
Day 8–9	Pilot: one department or location, live, reviewed daily	Both
Day 10–12	Full go-live, employee communication issued, HR team trained	Customer HR
Week 3+	Enable actions and conversational policy answers once the customer is comfortable	Both

The customer's checklist: a sample attendance export and employee master · the full list of attendance codes with meanings · holiday and shift calendars per location · manager mapping with contactable numbers · a WhatsApp business number and Meta access · their leave and regularisation policy (types, backdating windows, approvers) · HRMS access, read then write · an employee communication and consent note · a named HR owner and an IT contact.

14. Delivery roadmap

Five phases, each ending with something usable. Durations are indicative engineering elapsed time, assume the dependencies in section 15, and will be firmed up at planning.

PHASE	THEME	DELIVERED	INDICATIVE
0	Demo DONE	Single-customer proof: detection, WhatsApp question, AI reply reading, correct guidance, live dashboard	complete
1	Core platform, multi-customer	Customers, users and roles; employee master; manual and file ingestion; detection engine with calendars and exclusions; WhatsApp conversations with templates and languages; AI reply understanding; HR workspace; scheduling; audit trail; Azure environments and pipeline	8–10 weeks

2	Mapping studio and integrations	Field and code mapping with AI suggestions and versioning; dry-run sign-off tooling; generic file, REST and database connectors; scheduled imports with reject reporting; import health monitoring; operational reports and exports	6–8 weeks
3	Actions, approvals and dialogue	Action engine with pre-checks, idempotency and receipts; manager approval on WhatsApp with delegation and timeouts; apply and approve leave and regularisation; fallback write-back routes; policy pack and grounded policy answers; day-2 reminder logic	8–10 weeks
4	Conversational voice and depth	Day-3 calls with live speech conversation (Sarvam speech models, Plivo or Exotel), keypad fallback, transfer to HR, retries; escalation chains and SLA clocks; email and SMS fallbacks; manager digests; month-end and leadership analytics; AI accuracy tooling	8–10 weeks
5	Commercial readiness and scale	Usage metering and billing exports; white-labelling; named HRMS connectors and a profile library; self-service onboarding; performance hardening and penetration-test remediation	6–8 weeks

First revenue-ready release: end of Phase 2, with guidance-only messaging on real customer data. Phase 3 turns it from a chaser into an agent; Phase 4 adds the voice conversation. Phases can overlap where the team allows, and the order can change if a first customer's needs demand it — for example bringing a specific HRMS connector forward. Each phase ends with working software behind per-customer switches, a demo on real sample data, updated documentation, and automated tests for the checks that matter (no duplicate messages, no message from an unmapped code, no action without approval).

15. Assumptions, exclusions and risks

15.1 Assumptions and dependencies

Employees have WhatsApp on a mobile number the employer holds and may use for work · attendance data is available at least daily and is reasonably accurate (the agent explains gaps, it does not fix bad data) · each customer can state their attendance codes, leave types and approvers, and provide their policy in writing · managers are contactable on WhatsApp · customers accept WhatsApp, voice and AI usage charges as pass-through or bundled · India-first for launch: Indian numbers, Indian leave conventions, English, Hindi and Gujarati first. **Outside our control:** Meta template approvals, number verification and policy changes · the HRMS vendor's API and how fast access is granted · the customer's IT and HR providing samples, credentials, calendars and the employee communication · voice provider account approval and Indian calling clearances · model provider availability and pricing.

15.2 Explicitly out of scope

Replacing the HRMS or becoming the system of record · payroll processing or statutory filings · biometric devices and access-control hardware · shift rostering and workforce planning · performance management or disciplinary automation, and any decision affecting a person's employment (the agent gathers and records; people decide) · recruitment and unrelated HR workflows · a native employee mobile app (WhatsApp is the interface by design) · general HR chatbot duties beyond attendance, leave and regularisation.

15.3 Risks and how we handle them

RISK	HANDLING
WhatsApp template rejected, or Meta policy changes	Factual utility-category wording per customer; template status tracked; alternates pre-drafted; email and SMS fallback so a case never stalls
Customer's HRMS has no write API	Four fallback routes (5.3); valuable in guidance-only mode; actions enabled when access allows
Wrong person messaged through a mapping error	Unmapped codes never message; mandatory dry-run sign-off; pilot on one department; per-customer sending switch; every send traceable to its import row

AI misreads a reply, or answers a policy question wrongly	Buttons and numbers bypass AI; confidence thresholds with handover; answers grounded in the policy pack with the source shown; HR reviews samples before conversational mode is enabled; accuracy reported
Voice quality on poor lines or heavy accents	Indian-language speech models; keypad fallback always available; two-turn limit then HR; transcripts reviewed during the pilot
Employees find it intrusive	One message per gap, never broadcasts; quiet hours; clear opt-out; employee communication at launch; wording approved by the customer
Managers ignore approval requests	Reminder, then timeout escalation to skip-level or HR; delegation while on leave; pending-with-manager reporting
Bad or late attendance data	Import validation with reject reporting; catch-up runs; corrections close cases automatically; alerts when an expected import does not arrive
Cost surprises from messaging, calls or AI	Per-customer metering and caps; rules-first classification; small models; cached policy answers; spike alerts
Scope creep from each new customer's wishes	Customer-specific behaviour belongs in configuration, never in code; genuinely new needs become catalogued actions or connectors every customer can use

16. Running costs and pricing levers

COST DRIVER	BEHAVIOUR
WhatsApp conversations (Meta)	Charged per conversation, not per message, at Meta's country and category rates. Scales with the number of attendance gaps, not headcount. Follow-ups inside 24 hours are free — the flow is designed around that.
Voice calls	Per minute (Plivo or Exotel), plus speech recognition and synthesis per minute. Only for cases still unresolved on day 3 — a small fraction of cases.
AI models	Only for replies that are not buttons or numbers, and for policy questions. Small models, short prompts, cached answers: a fraction of a rupee per interaction at current rates.
Azure	App Service, PostgreSQL, Functions, Service Bus, Blob, Key Vault, monitoring. Largely fixed, shared across customers, stepping up with volume.
Onboarding effort	People time for mapping, integration and sign-off. Falls sharply once a profile exists for a given HRMS.
Support and success	Monitoring health, connector breakages after customer HRMS upgrades, template and policy changes.

Pricing models this supports: per employee per month, tiered by headcount, with messaging bundled under fair use or passed through · platform fee plus metered usage · per resolved case, which ties most closely to value and makes a strong pilot offer · plus a one-time onboarding fee, and separate pricing for named HRMS connectors that need new development. The platform meters everything needed for any of these from Phase 5, and reports volumes per customer from Phase 2, so pricing decisions rest on real numbers.

Appendix A — Attendance model and action catalogue

A.1 Canonical attendance meanings

The fixed internal vocabulary every customer's codes map into, so the engine and reports stay consistent: present · absent full day · absent first half · absent second half · absent half-day unspecified · missed or incomplete punch · short hours · late in / early out · leave approved · leave pending approval · leave rejected · leave without pay · on duty, work from home, travel, training · compensatory off · weekly off · holiday · optional holiday · suspension · long leave · not employed on that date · unknown (unmapped). **Chased by default:** the absence, half-day and missed-punch

meanings. **Never chased:** weekly off, holiday, not-employed, unknown. **Configurable:** short hours, lateness, leave without pay. **Leave pending approval** chases the manager rather than the employee.

A.2 Action catalogue v1

ACTION	PRE-CHECKS	RECORDED
Apply leave	Type allowed; balance available; inside the backdating window; no overlap; employee active on that date	HRMS reference, type, dates, approver, approval time
Apply regularisation	Date eligible; monthly limit not exceeded; reason captured	HRMS reference, date, reason, approver
Apply unpaid leave	Paid balance exhausted or explicit employee choice; HR approval where configured	HRMS reference, days, approvals
Approve / reject leave or regularisation	Approver authorised for that employee; request still pending	Decision, who, when, any note
Cancel or correct a request	Cancellable; inside the allowed window	Cancellation reference, reason
Raise HR ticket	Category available; description captured	Ticket reference, assignee
Nudge manager	Pending beyond configured hours; nudge limit not exceeded	Nudge count, channel, timestamps
Close case with reason	HR permission	Reason, user, timestamp

Later candidates, following the same pattern: shift-change requests, compensatory-off claims, work-from-home requests, document upload, overtime confirmation.

Appendix B — Message and conversation catalogue

Wording from the agreed flowchart, adaptable per customer within WhatsApp's template rules. Braces are filled per employee.

Stage 1 — the question (template)

Hi {employee_name}, our attendance record shows you are marked absent on {absent_date}. Please confirm the reason by choosing one option below.
Buttons: 1. Yes, I was absent · 2. No, I was working · 3. I have already applied leave · 4. I have already sent a regularisation request

Stage 1 — the replies

EMPLOYEE SAYS	AGENT REPLIES
1 — I was absent	Please apply leave if leave balance is available. If not, follow the HR process for unpaid leave or regularisation and get it approved by your manager. <i>(With actions on: offers to apply it.)</i>

2 — I was working	Please apply attendance regularisation for {absent_date} and get it approved by your manager. <i>(With actions on: offers to apply it.)</i>
3 — Already applied leave	Please ask your manager to approve your leave for {absent_date}. <i>(With actions on: nudges the manager.)</i>
4 — Already sent regularisation	Please ask your manager to approve your regularisation request for {absent_date}. <i>(With actions on: nudges the manager.)</i>
A question	Answered from the policy pack and live HRMS data (section 6.2), then back to the pending action
Unclear	One clarifying question; if still unclear, handed to HR and the employee is told a colleague will follow up

Stage 2 — the day-2 reminder (template, worded by the original answer)

ORIGINAL ANSWER	REMINDER
Absent / unpaid leave	Reminder: have you completed the leave or unpaid leave process for {absent_date} and got it approved by your manager? [Done] [Not done] [Need help]
Regularisation	Reminder: have you submitted regularisation for {absent_date} and got it approved by your manager? [Done] [Not done] [Need help]
Already applied leave	Reminder: has your manager approved your leave for {absent_date}? [Approved] [Not approved] [Need help]
Already sent regularisation	Reminder: has your manager approved your regularisation request for {absent_date}? [Approved] [Not approved] [Need help]
Never replied	The original question is sent again

Stage 3 — the day-3 call (spoken, not a recording)

Agent: "Hello {employee_name}, this is an automated HR assistant from {company}. Your attendance action for {absent_date} is still pending. Have you completed the leave or regularisation process?"

Employee: "I applied leave yesterday but my manager hasn't approved it."

Agent: "Thank you. I can see the request pending with {manager_name}. I'll remind them today and update HR. Anything else about your attendance?"

Keypad fallback: press 1 if completed, 2 if not, 3 to be contacted by HR.

Outcomes: completed → verified against the HRMS and closed · not completed → HR escalation · needs help → HR follow-up · no answer → retry, then HR escalation.

Manager messages (with actions enabled)

SITUATION	MESSAGE
Approval request	{employee_name} has requested {action} for {date}. Reason: {reason}. [Approve] [Reject] [Ask for details]
Nudge	Reminder: {employee_name}'s {action} for {date} is waiting for your approval.
Daily digest	{n} attendance items are pending with you. Reply to see them one by one.

NEXT STEP

Review this scope and mark anything to add, remove or re-sequence — particularly the action catalogue (A.2), the escalation timings, the default model per language, and which HRMS products the first customers use, since that decides which connectors are built first. On sign-off, Phase 1 planning begins with a detailed schedule and a fixed definition of done per module.

